

Factor Model: Identification and Consistency

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1 Factor Model

Consider a factor model

$$\begin{aligned}x_{ti} &= c_{ti} + \varepsilon_{ti} \\ &= f_t' \alpha_i + \varepsilon_{ti}\end{aligned}$$

for $i = 1, \dots, N$ and $t = 1, \dots, T$ where c_{ti} is the common component that admits a factor structure $c_{ti} = f_t' \alpha_i$ with

$$f_t = \begin{pmatrix} f_{t1} \\ \vdots \\ f_{tK} \end{pmatrix}, \alpha_i = \begin{pmatrix} \alpha_{i1} \\ \vdots \\ \alpha_{iK} \end{pmatrix}$$

Each of the observables x_{ti} is generated by three random elements f_t, α_i and ε_{ti} . We consider the conditions under which the common component $c_{ti} = f_t' \alpha_i$ is identified from the idiosyncratic component ε_{ti} .

1.1 Identification of Common Component

In its matrix form, the factor model can be written as

$$X = FA' + E$$

where X, E are $T \times N$ dimensional matrices, F and A are $T \times K$ and $N \times K$ dimensional matrices respectively. Consider the singular value decomposition of the common component and the idiosyncratic component.

$$\begin{aligned}FA' &= U^{(c)} D^{(c)} V^{(c)'} \\ E &= U^{(id)} D^{(id)} V^{(id)'}\end{aligned}$$

where $U^{(c)}$ and $V^{(c)}$ are $T \times K$ and $N \times K$ dimensional matrices, $U^{(id)}$ and $V^{(id)}$ are $T \times L$ and $N \times L$ with $L = \min(N, T)$, and $D^{(c)}, D^{(id)}$ are $K \times K$, $L \times L$ dimensional diagonal matrices. Under the conditions

$$\frac{1}{T} F' F = \frac{1}{T} \sum_{t=1}^T f_t f_t' \rightarrow_p M_F > 0 \quad (1)$$

$$\frac{1}{N} A' A = \frac{1}{N} \sum_{i=1}^N \alpha_i \alpha_i' \rightarrow_p M_A > 0 \quad (2)$$

the Frobenius norm of the common component is of the order

$$\begin{aligned}\|FA'\|_F^2 &= \text{tr}(FA'AF') \\ &= \text{tr}(A'AF'F) \\ &= O_p(NT)\end{aligned}$$

This implies that

$$\begin{aligned}\|FA'\|_F^2 &= \text{tr}(FA'AF') \\ &= \text{tr}\left(U^{(c)}D^{(c)}V^{(c)'}V^{(c)}D^{(c)}U^{(c)'}\right) \\ &= \text{tr}\left(D^{(c)2}\right) \\ &= \sum_{j=1}^K \delta_j^{(c)2} \\ &= O_p(NT)\end{aligned}$$

where $(\delta_j^{(c)})_{j=1}^K$ are singular values of the common component. Since both M_F and M_A are assumed to be positive definite and K is finite, the squared singular values are all $O_p(NT)$. This implies that $(\delta_j^{(c)})_{j=1}^K$ are all $O_p(\sqrt{NT})$.

Now, consider the Frobenius norm of the idiosyncratic component E . As long as (e_{ti}) have positive variances,

$$\begin{aligned}\|E\|_F^2 &= \text{tr}(E'E) \\ &= \sum_{i=1}^N \sum_{t=1}^T e_{ti}^2 \\ &= O_p(NT)\end{aligned}$$

which is of the same order as the Frobenius norm of the common component. Let $(\delta_j^{(ic)})_{j=1}^{\min(N,T)}$ be the singular values of E . Then, we have $\sum_{j=1}^{\min(N,T)} \delta_j^{(ic)2} = O_p(NT)$. We want the order of $(\delta_j^{(ic)})$ to be smaller than $O_p(\sqrt{NT})$ so that the common component is identified from the idiosyncratic component. Suppose $(\delta_j^{(ic)})$ all have the same order. Then, we have

$$\min(N, T) \delta_j^{(ic)2} = O_p(NT)$$

for all $j = 1, \dots, \min(N, T)$. In other words, the operator norm of E is given by

$$\|E\|_{op} = \max_{j=1, \dots, \min(N, T)} \delta_j = O_p(\sqrt{\max(N, T)}) \quad (3)$$

The condition (3) implies that the rates of the singular values of FA' and that of E are different and thus are identified from each other as both N and T diverge. Under what restrictions on (e_{ti}) would this be achieved? From random matrix theory, if (e_{ti}) are independent across i and t and have finite fourth moments, we have (3) satisfied.¹ To what

¹E.g., Latala, R. (2005). "Some estimates of norms of random matrices"

extent can we relax the assumption that (e_{ti}) are independent? Suppose the error matrix E can be written as

$$E = \Gamma^{1/2} U \Sigma^{1/2}$$

for some $T \times T$ and $N \times N$ dimensional positive definite non-random matrices Γ and Σ and $T \times N$ dimensional random matrix U with independent elements. Suppose further that each element of U satisfies $\mathbb{E}u_{ti} = 0$, $\mathbb{E}u_{ti}^2 = 1$ and $\mathbb{E}u_{ti}^4 < \infty$ where u_{ti} are (t, i) elements of U . This implies that e_{ti} is given by

$$e_{ti} = \sum_{s=1}^T \sum_{j=1}^N \gamma_{ts} u_{sj} \sigma_{ji}$$

where γ_{ts} and σ_{ji} are (t, s) and (j, i) elements of $\Gamma^{1/2}$ and $\Sigma^{1/2}$ respectively. Suppose further that

$$\begin{aligned} \|\Sigma\|_{op} &< \infty \\ \|\Gamma\|_{op} &< \infty \end{aligned}$$

This implies that

$$\|E\|_{op} < \|\Gamma^{1/2}\|_{op} \|U\|_{op} \|\Sigma^{1/2}\|_{op} = O_p(\sqrt{\max(N, T)})$$

in particular, because the operator norm is sub-multiplicative and elements of U are independent and have finite fourth moments. This implies that the error term E may have a limited amount of both cross-sectional and serial correlation and still have $\|E\|_{op} = O_p(\sqrt{\max(N, T)})$.

The condition (3) also implies the 'non-pervasiveness' of the idiosyncratic component. In other words, under the condition (3), the idiosyncratic error e_{ti} does not admit a representation

$$e_{ti} = \tilde{f}_t' \tilde{\alpha}_i \tag{4}$$

where both $\tilde{f}_t$ and $\tilde{\alpha}_i$ does not vanish elementwise in the limit. To see this, consider again the singular value decomposition

$$\begin{aligned} E &= U^{(id)} D^{(id)} V^{(id)'} \\ &= (\sqrt{T} U^{(id)}) \left(\frac{1}{\sqrt{NT}} D^{(id)} \right) (\sqrt{N} V^{(id)'}) \end{aligned}$$

The (t, i) -th element of E is given by

$$e_{ti} = (\sqrt{T} U_{t, \bullet}^{(id)'}) \left(\frac{1}{\sqrt{NT}} D^{(id)} \right) (\sqrt{N} V_{i, \bullet}^{(id)})$$

where $U_{t, \bullet}^{(id)}$ and $V_{i, \bullet}^{(id)}$ are t -th and i -th row of $U^{(id)}$ and $V^{(id)}$ respectively. Here, both $\sqrt{T} U_{t, \bullet}^{(id)}$ and $\sqrt{N} V_{i, \bullet}^{(id)}$ have elements that are bounded as $N, T \rightarrow \infty$. However, each of the diagonal elements of $\frac{1}{\sqrt{NT}} D^{(id)}$ are $O_p\left(\frac{1}{\sqrt{\min(N, T)}}\right)$ so that in the representation (4), either $\tilde{f}_t$ or $\tilde{\alpha}_i$ have elements that are vanishing in the limit. On the other hand, it is clear that the conditions (1) and (2) are sufficient conditions for the pervasiveness of the factors (f_t) and the loadings (α_i) .

1.2 Identification of the Factors

Even if the common component $c_{ti} = f'_t \alpha_i$ is identified from the idiosyncratic error ε_{ti} , the factors f_t are still not identified from the loadings α_i . This is because we may modify f_t as Rf_t and α_i as $R'^{-1}\alpha_i$ for any invertible matrix R . To identify the factors f_t , we may impose

$$\begin{aligned} \frac{1}{T} \sum_{t=1}^T f_t f'_t &\rightarrow_p I_K \\ \frac{1}{N} \sum_{i=1}^N \alpha_i \alpha'_i &\rightarrow_p D \end{aligned} \tag{5}$$

where D is a diagonal matrix with distinct decreasing elements.

1.3 Estimation

Consider the singular value decomposition of X

$$\begin{aligned} X &= UDV' \\ &= U_{1:K} D_{1:K} V'_{1:K} + U_{K+1:\bullet} D_{K+1:\bullet} V'_{K+1:\bullet} \end{aligned}$$

where we have written the leading K elements of the singular value decomposition separately. Since the singular values of the common component is of higher order, we naturally want to use $U_{1:K} D_{1:K} V'_{1:K}$ as estimates for the common component. With the identification restriction (5), we may use

$$\begin{aligned} \hat{F} &= \sqrt{T} U_{1:K} \\ \hat{A} &= \sqrt{N} V_{1:K} \left(\frac{1}{\sqrt{NT}} D_{1:K} \right) \end{aligned}$$

as our estimates. We see that our estimates satisfies $\hat{F}'\hat{E} = 0$ and $\hat{E}\hat{A} = 0$ by design where $\hat{E}$ is the $T \times N$ dimensional matrix of residuals. This implies we need to assume further that

$$\frac{1}{T} \sum_{t=1}^T f_t e_{ti} = O_p \left(\frac{1}{\sqrt{T}} \right) \tag{6}$$

$$\frac{1}{N} \sum_{i=1}^N \alpha_i e_{ti} = O_p \left(\frac{1}{\sqrt{N}} \right) \tag{7}$$

as $T, N \rightarrow \infty$ for all i and t .

1.4 Theory: Consistency

² From $X = FA' + E$, we have

$$\frac{1}{NT} XX' = \frac{FA' A F'}{N} \frac{1}{T} + \frac{FA' E'}{NT} + \frac{EAF'}{NT} + \frac{EE'}{NT}$$

²We mainly follow Bai and Ng (2023) for the proof of consistency.

From the definition of $\hat{F}$ and $XX' = UD^2U'$, we have

$$\begin{aligned}\frac{1}{NT}XX'\hat{F} &= \frac{1}{NT}UD^2U'(\sqrt{T}U_{1:K}) \\ &= \frac{1}{NT}(\sqrt{T}U_{1:K})D_{1:K}^2 \\ &= \frac{1}{NT}\hat{F}D_{1:K}^2\end{aligned}$$

so that we have

$$\left(\frac{FA'AF'}{N} \frac{1}{T} + \frac{FA'E'}{NT} + \frac{EAF'}{NT} + \frac{EE'}{NT}\right)\hat{F} = \frac{1}{NT}\hat{F}D_{1:K}^2$$

Rearranging, we have

$$\hat{F} - FH_{NT} = \left(\frac{FA'E'\hat{F}}{NT} + \frac{EAF'\hat{F}}{NT} + \frac{EE'\hat{F}}{NT}\right)\left(\frac{1}{NT}D_{1:K}^2\right)^{-1}$$

where

$$H_{NT} = \left(\frac{A'AF'\hat{F}}{NT}\left(\frac{1}{NT}D_{1:K}^2\right)^{-1}\right)$$

This implies

$$\begin{aligned}\frac{1}{\sqrt{T}}\|\hat{F} - FH_{NT}\|_F &= \frac{1}{\sqrt{T}}\left\|\left(\frac{FA'E'\hat{F}}{NT} + \frac{EAF'\hat{F}}{NT} + \frac{EE'\hat{F}}{NT}\right)\left(\frac{1}{NT}D_{1:K}^2\right)^{-1}\right\|_F \\ &< \frac{1}{\sqrt{T}}\left(\left\|\frac{FA'E'\hat{F}}{NT}\right\|_F + \left\|\frac{EAF'\hat{F}}{NT}\right\|_F + \left\|\frac{EE'\hat{F}}{NT}\right\|_F\right)\left\|\left(\frac{1}{NT}D_{1:K}^2\right)^{-1}\right\|_F \\ &< \left[2\left(\frac{\|F\|_F\|\hat{F}\|_F}{T}\right)\frac{\|A'E'\|_F}{\sqrt{TN}} + \frac{\|EE'\hat{F}\|_F}{NT^{3/2}}\right]\left\|\left(\frac{1}{NT}D_{1:K}^2\right)^{-1}\right\|_F\end{aligned}$$

Here, we have

$$\frac{\|F\|_F\|\hat{F}\|_F}{T} = O_p(1)$$

by (1) and definition of $\hat{F}$. Also,

$$\frac{\|EE'\hat{F}\|_F}{NT^{3/2}} < \frac{\|E\|_{op}^2\|\hat{F}\|_F}{NT^{3/2}} = O_p\left(\frac{\max(N, T)}{NT}\right) = O_p\left(\frac{1}{N}\right) + O_p\left(\frac{1}{T}\right)$$

where we've used (3). Using $A'E' = \sum_{i=1}^N \alpha_i e'_{\bullet, i}$, we have

$$\begin{aligned}\|A'E'\|_F^2 &= \sum_{k=1}^K \sum_{t=1}^T \left(\sum_{i=1}^N \alpha_{ik} e_{ti}\right)^2 \\ &= N^2 \sum_{k=1}^K \sum_{t=1}^T \left(\frac{1}{N} \sum_{i=1}^N \alpha_{ik} e_{ti}\right)^2 \\ &= O_p(NT)\end{aligned}$$

due to (7). This implies

$$\frac{\|A'E'\|_F}{\sqrt{TN}} = O_p\left(\frac{1}{\sqrt{N}}\right)$$

Finally, note that $\frac{1}{NT}D_{1:K}^2$ is K largest eigenvalues of $\frac{1}{NT}XX'$. In the expansion

$$\frac{1}{NT}XX' = \frac{1}{NT}FA'AF' + \frac{1}{NT}FA'E' + \frac{1}{NT}EAF' + \frac{1}{NT}EE'$$

the largest eigenvalues of the three terms on the right hand side converge to zero as $N, T \rightarrow \infty$ due to (6), (7), and (3). By the matrix perturbation theorem, this implies that $\frac{1}{NT}D_{1:K}^2$ converges to the largest K eigenvalues of $\frac{1}{NT}FA'AF'$ as $N, T \rightarrow \infty$. Since the latter is equal to the largest K eigenvalues of $\frac{1}{NT}A'AF'F$, we have

$$\frac{1}{NT}D_{1:K}^2 = O_p(1)$$

Collecting all the results so far, we have

$$\frac{1}{\sqrt{T}}\|\hat{F} - FH_{NT}\|_F = O_p\left(\frac{1}{\sqrt{N}}\right) + O_p\left(\frac{1}{T}\right)$$

which gives

$$\frac{1}{T}\|\hat{F} - FH_{NT}\|_F^2 = \frac{1}{T}\sum_{t=1}^T\|\hat{f}_t - H'_{NT}f_t\|^2 = O_p\left(\frac{1}{N}\right) + O_p\left(\frac{1}{T^2}\right)$$

We may switch the role played by F and A to obtain

$$\frac{1}{N}\|\hat{A} - A(H'_{NT})^{-1}\|_F^2 = \frac{1}{N}\sum_{i=1}^N\|\hat{\alpha}_i - H_{NT}^{-1}\alpha_i\|^2 = O_p\left(\frac{1}{T}\right) + O_p\left(\frac{1}{N^2}\right)$$

1.5 Remarks

The assumptions on E play a key role in both identification of the common components and proof of consistency. We may relax the condition

$$\|E\|_{op} = O_p(\sqrt{\max(N, T)})$$

to any rate less than $O_p(\sqrt{NT})$ for the identification. Suppose $\|E\|_{op} = O_p(\sqrt{\delta_{NT}}) < O_p(\sqrt{NT})$. Then, the convergence rate of the factor becomes

$$\frac{1}{T}\|\hat{F} - FH_{NT}\|_F^2 = O_p\left(\frac{1}{N}\right) + O_p\left(\frac{\delta_{NT}^2}{N^2T^2}\right)$$